

Bed frames, support and mattress fit

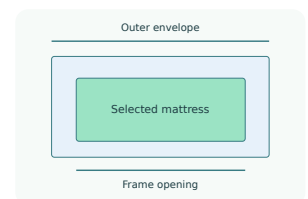
Mattress, frame and support form one coordinated system. Distinguish fit geometry from load-carrying performance and identify the information needed before releasing a bed-base detail.

What you will accomplish

- Match the mattress and frame opening
- Trace the support and load path
- Review assembly, service and verification

Match the mattress and frame opening

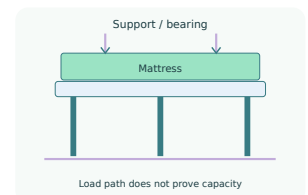
Record the mattress model, dimensions and support requirements, then show frame opening, external envelope and sleeping level. A commercial size name does not guarantee interchangeability. Check mattress placement/removal and bedding access with the selected products. Interface clearance comes from their requirements and the detail, not a universal extra dimension added to every bed.



Defined mattress, frame opening and final sleeping level.

Trace the support and load path

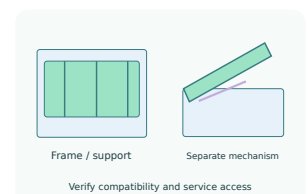
Trace loads from mattress to foundation, frame and floor supports. Show how any centre support connects and bears, rather than drawing an unsupported line. Slats and a solid deck are not automatically equivalent: check mattress compatibility, ventilation and performance. Slat dimensions, spacing and support count require verified system information, not inference from a picture.



Every support shown needs a defined connection and bearing path.

Review assembly, service and verification

Define the frame/support/mattress assembly sequence and connector access. A lift-up storage base adds a mechanism with its own compatibility, restraint and movement requirements; it must not be attached to an unverified frame by guesswork. Inspect the assembled system using approved requirements and record results and use instructions rather than inventing a load test.



A lifting mechanism is a separate system decision.

Worked case and decision

Option A replaces slats with a solid deck without checking mattress requirements. Option B documents compatible support, bearing and connections. Develop B, or obtain a technical review of A's change before accepting it. A simpler appearance does not prove compatibility.

Decision: verify mattress compatibility and performance before changing support type.

Check understanding

Can a simpler drawing justify replacing slats with a solid deck?

- No; verify mattress, performance and ventilation requirements
- Yes; all foundations are equivalent
- Yes, if the board looks thick

Correct: No; verify mattress, performance and ventilation requirements. The change affects the complete system; appearance does not demonstrate suitability.

Lesson questions

Does a slat drawing prove bed capacity?

No. Materials, system, connections and performance need verification.

Practice

Draw a base section and load path, with missing information required for approval.

Mattress and opening

Support and connections

Assembly and verification

Review your work against the criteria. Saving it does not constitute technical project approval.

- No universal clearance
- Defined bearing path
- No invented capacity/test